

Parameters and architecture

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can distinguish architectural choices from parameter values.

I can count weights and biases in a small dense network.

I can explain why parameter count alone cannot establish quality.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Representing inputs as vectors. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Distinguish architectural choices from parameter values

Count weights and biases in a small dense network

Explain why parameter count alone cannot establish quality

Retrieve what you already know

In the Week 6 network, which quantities change when x changes while the model stays fixed?

My starting explanation

This week's end goal

By the end, I can explain parameters and architecture using a traced example and a stated limitation.

2. Notice the evidence

Consider a dense network with 2 inputs, 3 hidden units using ReLU, and 1 output unit using identity. We call its widths 2 to 3 to 1. These widths, connectivity and activation choices describe its architecture. The weights and biases supply the numerical values used by that architecture.

A fully connected hidden unit receiving 2 inputs needs 2 weights plus 1 bias. Three such units need 6 weights and 3 biases. The output receives 3 activations and needs 3 weights plus 1 bias. Total: 13 parameters. Input values and computed activations are not additional parameters.

Layer · Weights · Biases · Total

Layer: 2 inputs to 3 hidden units

Weights: 2 times 3 = 6

Biases: 3

Total: 9

Layer: 3 hidden values to 1 output

Weights: 3 times 1 = 3

Biases: 1

Total: 4

Layer: Whole network

Weights: 9

Biases: 4

Total: 13

This counting rule assumes one separate bias per output unit and no shared or tied weights. Other architectures can use different rules.

List the given facts and one fact that is not supplied.

3. See how it works

For a dense layer with n input components and m output units, each output has n weights and one bias. The count is n times m plus m , or $(n + 1)$ times m . Count each layer separately before adding.

```
def dense_count(inputs, outputs):  
    return inputs * outputs + outputs  
  
print(dense_count(2, 3))    # 9  
print(dense_count(3, 1))    # 4  
print(dense_count(2, 3) + dense_count(3, 1))    # 13
```

The helper assumes positive integer widths and the bias convention stated above. It is a counting aid, not a model trainer. A ReLU has no fitted numerical parameters in this example.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Classify hidden activation 0.7, weight -2, hidden width 3 and learning rate 0.1 as computed value, parameter, architecture choice or training setting.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Increase hidden width from 3 to 4.

Prediction: First layer 12 parameters; output layer 5; total 17.

Reason: Each new hidden unit adds 2 input weights, 1 hidden bias and 1 outgoing weight.

5. Words that unlock the idea

Word · Meaning

Word: Parameter

Meaning: A numerical model value that a fitting procedure may adjust

Word: Architecture

Meaning: The arrangement of layers, connections and activation choices

Word: Dense layer

Meaning: A layer in which each output unit has a weight for every input component

Word: Hyperparameter

Meaning: A setting chosen for a model or training procedure rather than fitted as an ordinary weight

Use two terms to explain a step in this week's example.

6. Follow a worked example

Decide what a change alters

Proposed change · Type of change · Consequence

Proposed change: Change one weight 2 to 3

Type of change: Parameter value

Consequence: Same architecture

Proposed change: Use 4 hidden units

Type of change: Architecture width

Consequence: Different parameter count

Proposed change: Use learning rate 0.1

Type of change: Training setting

Consequence: No extra model weight

Each row is a separate proposed change. A parameter value, an architectural choice and a training setting answer different questions about the system.

For a dense 2 to 4 to 1 network with one bias per unit, count each layer and the total. How many more parameters does it have than 2 to 3 to 1?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Parameters and architecture

Count trainable weights and biases separately from activation functions.

The facts we will use

Dense toy architecture 2 inputs → 3 hidden units → 1 output. Every hidden and output unit has one bias; hidden activation is ReLU.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Trainable parameter count

	weights	biases	total
input to hidden	6	3	9
hidden to output	3	1	4

Read a cell using both its row and column labels.

Parameters and architecture

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count first layer

$2 \times 3 = 6$ weights plus 3 biases gives 9 parameters.



2. Count output layer

$3 \times 1 = 3$ weights plus 1 bias gives 4 parameters.



3. Total

Total trainable parameters $= 9 + 4 = 13$; ReLU adds no parameters here.

CHANGE ONE THING

Increase hidden width from 3 to 4.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Count first layer

What happens and why: $2 \times 3 = 6$ weights plus 3 biases gives 9 parameters.

Step: 2. Count output layer

What happens and why: $3 \times 1 = 3$ weights plus 1 bias gives 4 parameters.

Step: 3. Total

What happens and why: Total trainable parameters=9+4=13; ReLU adds no parameters here.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Count one weight for each input-to-unit connection and one bias for each receiving unit. Two inputs connecting to three hidden units make six weights. The three hidden outputs connect to one output unit through three more weights. ReLU applies a fixed function and adds no adjustable parameter here. Architecture determines the connection pattern; learned values occupy those connections.

What this example does not establish

Parameter count describes capacity and storage structure, not accuracy, intelligence or guaranteed generalisation.

7. Find and repair a mistake

Two networks can share an architecture and still make different predictions because their parameter values differ. Conversely, two architectures can happen to produce identical outputs on the few cases tested. A count describes the model, not how well it has learned.

When comparing widths, keep the task, feature schema, training data, validation cases and evaluation rule fixed. Record the training budget and seed as well. More parameters can increase flexibility, but they can also increase resource use and overfitting risk. Neither outcome follows from counting alone.

Locate the earliest unsupported step

Recount the 2 to 3 to 1 model if every bias is omitted. State what the original count assumed.

State one repair and the checks you would rerun.

8. Practise together

A dense layer has 4 inputs and 2 outputs with one bias per output. Count weights, biases and total.

For a dense 2 to 4 to 1 network with one bias per unit, count each layer and the total. How many more parameters does it have than 2 to 3 to 1?

Explain your prediction to a partner before comparing results.

Practise with less help: Sort the quantities

Use workbook W2 for the connected example “Parameters and architecture”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Parameters and architecture. Proposed explanation: “Count one bias per input”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

How many new parameters did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Classify hidden activation 0.7, weight -2, hidden width 3 and learning rate 0.1 as computed value, parameter, architecture choice or training setting.

```
def dense_count(inputs, outputs):  
    return inputs * outputs + outputs  
  
print(dense_count(2, 3))    # 9  
print(dense_count(3, 1))    # 4  
print(dense_count(2, 3) + dense_count(3, 1))    # 13
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

A dense layer has 4 inputs and 2 outputs with one bias per output. Count weights, biases and total.

Recount the 2 to 3 to 1 model if every bias is omitted. State what the original count assumed.

A team says its 26-parameter model must outperform a 14-parameter model. Give a fair check and one reason the claim is unsupported.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled parameters and architecture evidence sheet to Image-pattern Network Investigation. Show how you can distinguish architectural choices from parameter values. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

A reproducible description needs ordered input features, layer widths, connections, activations and the exact parameter values or a way to recover them. A description of training also needs the data split and training settings. The label small neural network does not specify a computation.

Two teams both report 13 parameters. What would you request before claiming they built the same model?

Next week defines a loss so that we can measure how a particular set of parameters performs on labelled examples.

A team says its 26-parameter model must outperform a 14-parameter model. Give a fair check and one reason the claim is unsupported.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Decide what a change alters

Proposed change · Type of change · Consequence

Proposed change: Change one weight 2 to 3

Type of change: Parameter value

Consequence: Same architecture

Proposed change: Use 4 hidden units

Type of change: Architecture width

Consequence: Different parameter count

Proposed change: Use learning rate 0.1

Type of change: Training setting

Consequence: No extra model weight

Each row is a separate proposed change. A parameter value, an architectural choice and a training setting answer different questions about the system.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Count a new architecture independently

A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the output layer, with one bias per unit. Give each layer count and the total. If hidden width becomes 4, recount and explain why activation values are excluded.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the output layer, with one bias per unit. Give each layer count and the total. If hidden width becomes 4, recount and explain why activation values are excluded.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.